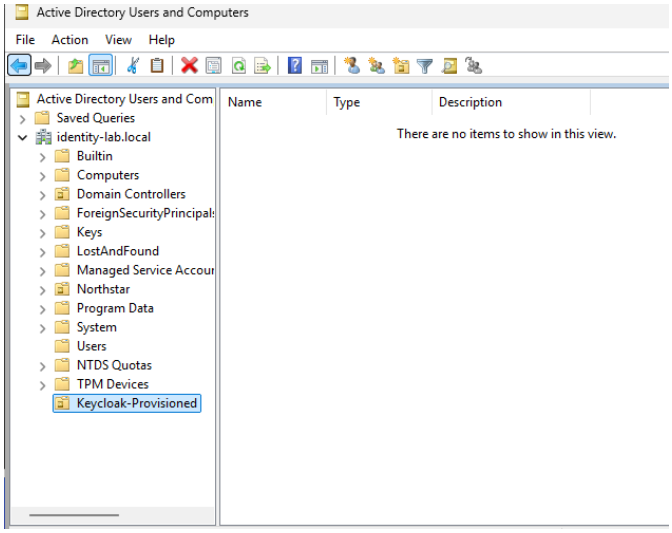
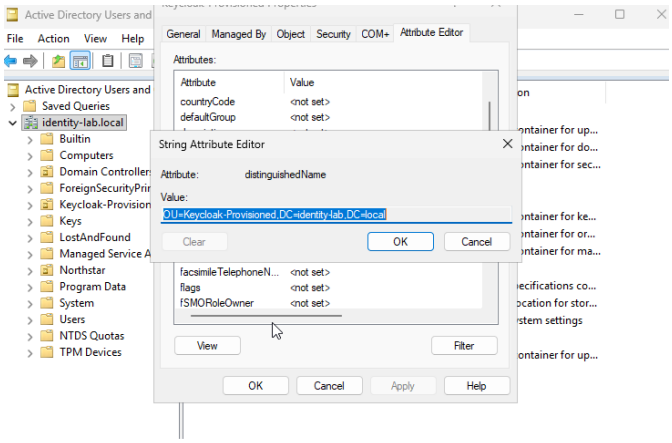
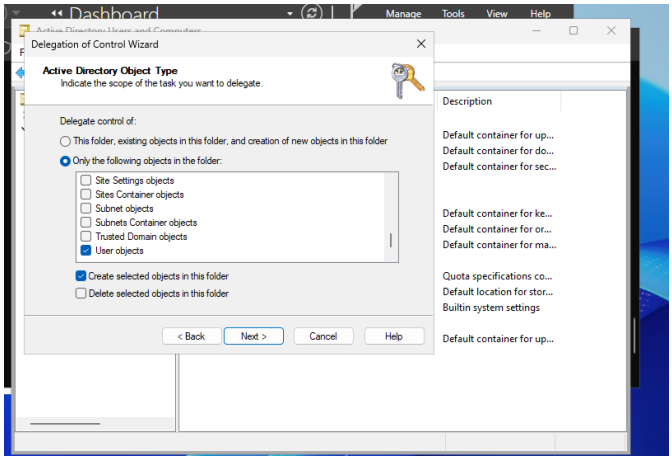
1. **Creating the Organizational Unit**
I created the Keycloak-Provisioned organizational unit in Active Directory to store accounts provisioned through Keycloak.



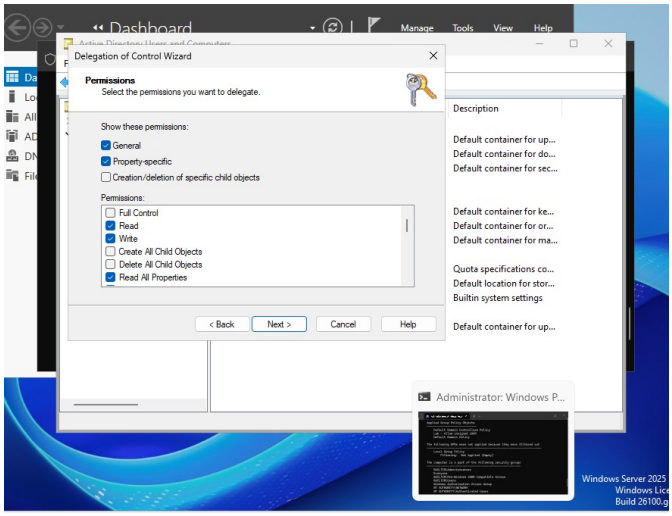
2. **Verifying the Distinguished Name**
I verified that the OU distinguished name was OU=Keycloak-Provisioned,DC=identity-lab,DC=local.



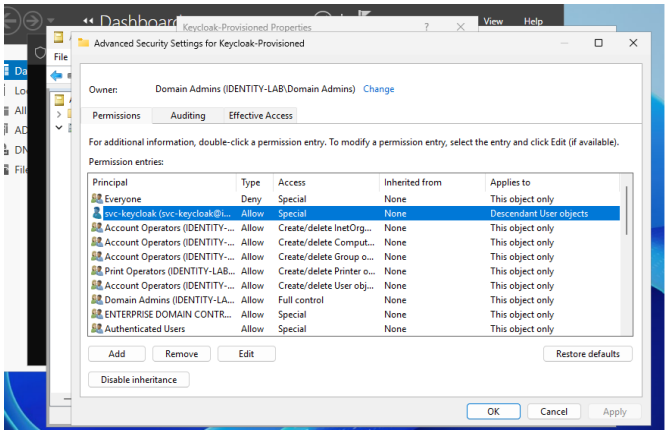
3. **Delegating User Creation Permissions**
I selected User objects and allowed the svc-keycloak account to create users within the designated OU without granting deletion permissions.



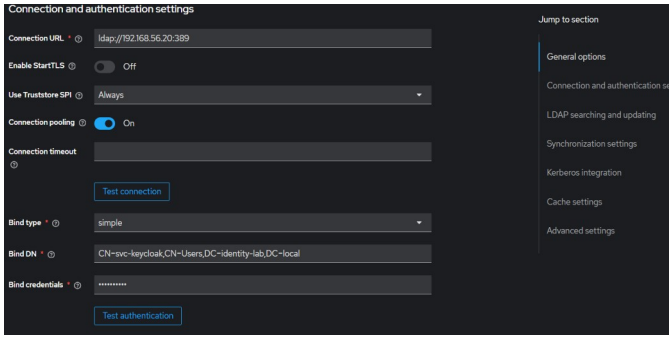
4. **Configuring Attribute Permissions**
I granted the service account read and write permissions for user properties within the provisioning OU.



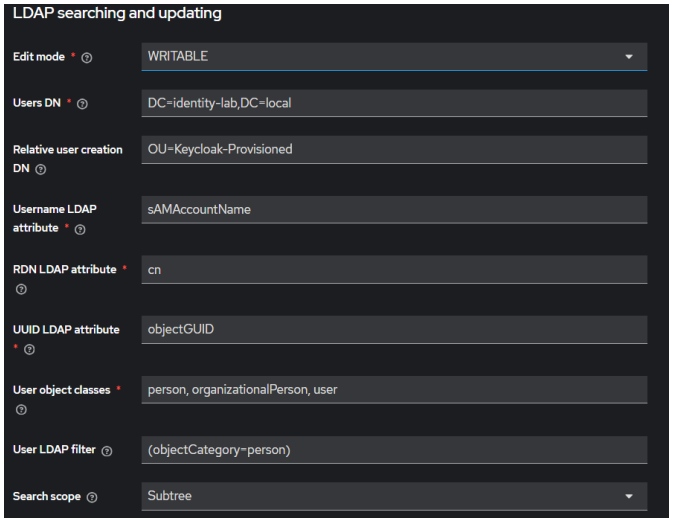
5. **Reviewing the Delegated Permissions**
I reviewed the OU security settings to verify that the delegation entries were assigned to svc-keycloak.



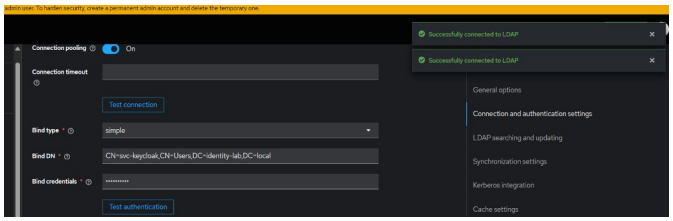
6. **Checking the LDAP Connection Settings**
I checked the LDAP connection URL and service account Bind DN used by Keycloak to access Active Directory.



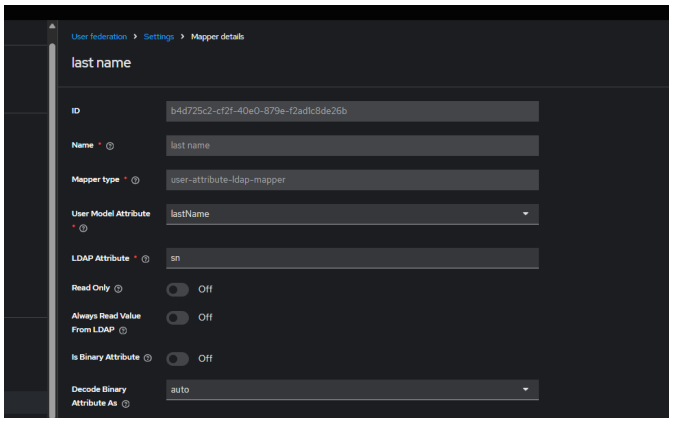
7. **Enabling LDAP Write Operations**
I configured the LDAP provider in WRITABLE mode with a domain-wide subtree search and Keycloak-Provisioned as the relative user creation DN.



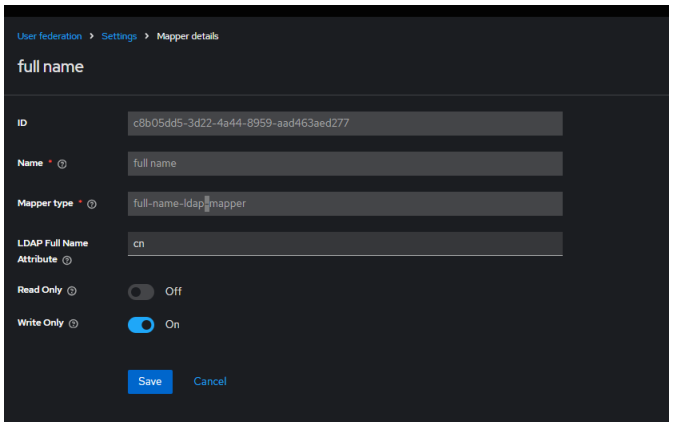
8. **Enabling User Provisioning**
I enabled Import users and Sync registrations so that new users created through Keycloak would be provisioned in Active Directory.



9. **Creating the First Name Mapper**
I manually added the missing first name mapper and configured it to write the Keycloak firstName value to the AD givenName attribute.

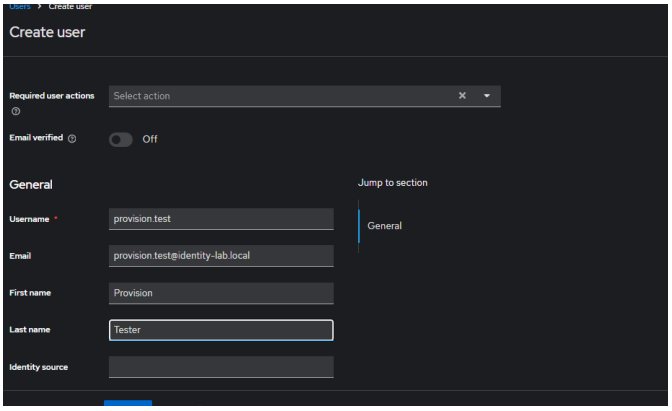


10. **Configuring the Full Name Mapper**
I configured the full name mapper to write the user's full name to displayName instead of cn.

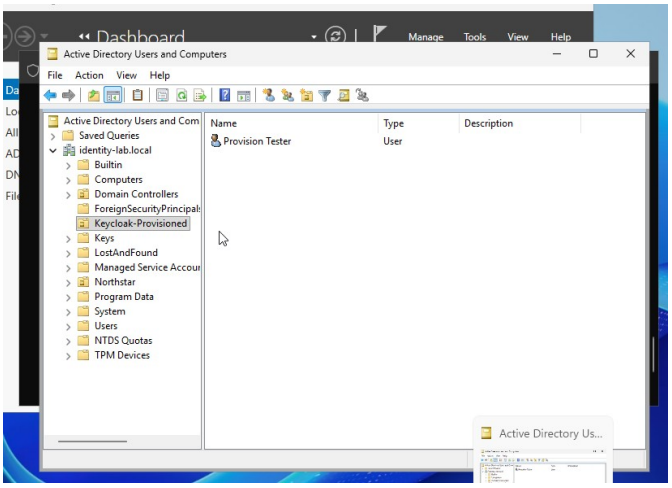


11. **Creating the Test User in Keycloak**
I created the provision.test user in Keycloak with

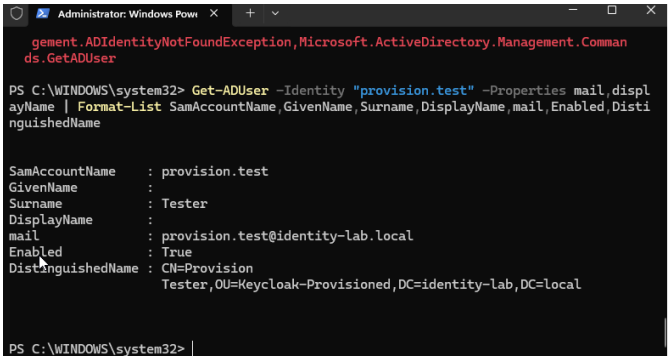
an initial first name, last name, and email address.



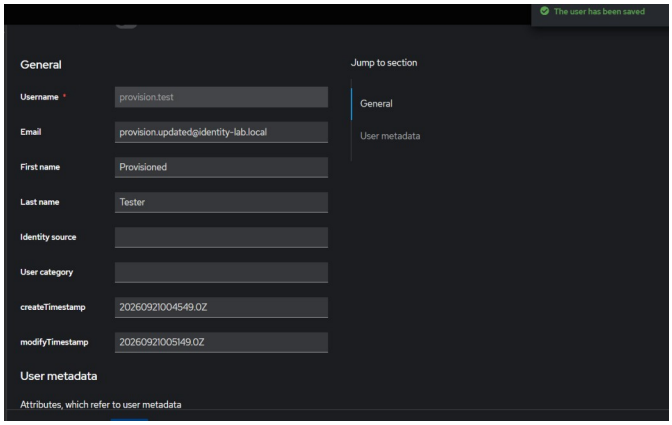
12. **Verifying Automatic User Creation in AD**
I verified that the user created in Keycloak appeared automatically in the Keycloak-Provisioned OU.



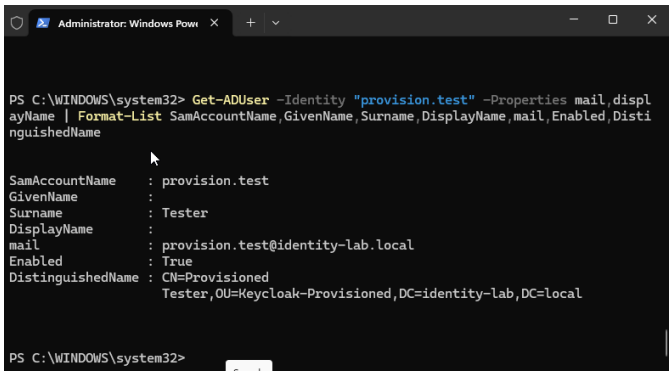
13. **Inspecting the Initial AD Attributes**
I used PowerShell to inspect the provisioned user's attributes, distinguished name, and initial disabled status.



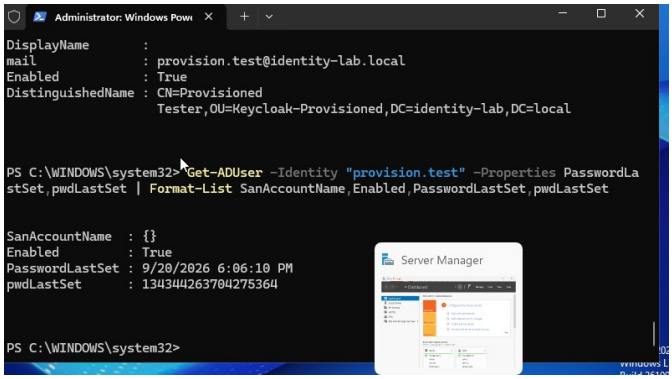
14. **Updating the User Profile in Keycloak**
I changed the user's first name to Provisioned and email address to provision.updated@identity-lab.local in Keycloak.



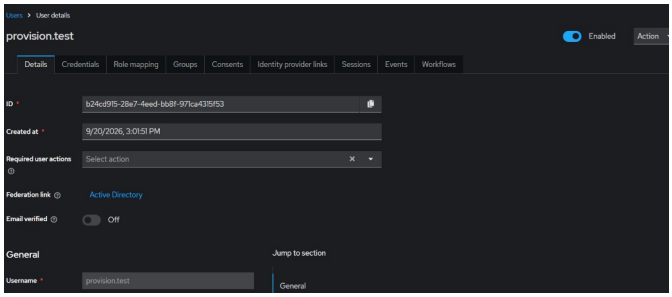
15. **Verifying Attribute Updates in AD**
I used PowerShell to verify that the updated first name, email address, and display name were written to Active Directory.



16. **Setting the Password and Enabling the AD Account**
I set the user's password directly in Active Directory and verified that the account was enabled.



17. **Synchronizing the Account Status with Keycloak**
I synchronized the LDAP user data and verified that provision.test was enabled in Keycloak.



18. **Testing Authentication**

I successfully signed in to the Keycloak Account Console as provision.test using the password set in Active Directory.

